At the short end of the yield curve, macro factors explain approximately 85% of the variation of yield levels decreasing to 40% of long-term bond yields. Of the macro factors, yield movements are most sensitive to inflation and inflation risk, which is not surprising because bonds are nominal securities. Monetary policy plays a large role in the "other factors," and monetary policy risk is priced in long-term bonds, but Ang and Piazzesi use other latent factors that may represent other risk factors.

Ang, Bekaert, and Wei (2008) decompose yields into real yields, expected inflation, and risk premiums based on the Fisher decomposition in equation (9.3). In their sample, they find that the short rate is 5.4% and the long yield is 6.3%, which can be decomposed as:

| | **Real Rate** | **Expected Inflation** | **Risk Premium** | **Total** |
|---|---|---|---|---|
| Short End | 1.2% | 3.9% | 0.3% | 5.4% |
| Long End | 1.3% | 3.9% | 1.1% | 6.3% |

Thus, most of the upward-sloping nominal yield curve (going from 5.4% to 6.3% in the Ang–Piazzesi sample) is explained by risk premiums rather than by real rates. The variance of the long yield is 20% attributable to variations of real rates, 70% due to movements in expected inflation, and 10% due to changes in risk premiums. Thus, expected inflation and inflation risk are extremely important determinants of long-term bond prices.

The risk premiums on long-term bonds also vary through time. Risk premiums are *counter-cyclical*: long-term bond yields are high relative to short rates during recessions because agents demand large compensation for taking on risk. The short rate is *pro-cyclical* because of the Taylor rule; the Fed lowers short rates in recessions to stimulate economic activity. During expansions, the opposite relations hold. Bonds do badly during recessions, where they have low prices and high yields. Investors feel most risk averse during these times of high marginal utility (see chapter 6). They demand higher risk premiums, which are reflected in a steeper yield curve. We tend to see steeper yield curves during other hard times like periods of high inflation, which tend to coincide with times of low economic activity. Bad times are also times of low productivity, and they could be times when other supply shocks hit the economy, like oil price shocks.[21]

The counter-cyclical risk premiums of long-term bonds manifest themselves in the ability of the term spread to predict recessions. Recessions have upward-sloping yield curves when the risk premiums on bonds are the highest. Correspondingly, the peaks of expansions are when risk premiums are the lowest,

[21] See Piazzesi and Schneider (2006), Wachter (2006), and Rudebusch and Swanson (2012).